FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Congenital Abnormalities & Down Syndrome

Tutorial | Week 5 · Measles, vaccines & the newborn | Paediatrics | Dr Aliku

Congenital anomalies affect about 1 in 33 babies; Down syndrome (trisomy 21, about 1 in 700–1000 births, rising steeply with maternal age) is the commonest chromosomal cause of intellectual disability and comes with predictable associated problems (heart, gut, thyroid, blood, eyes, ears, airway, neck) that a structured screening schedule catches early; Edith must also recognise and give first care for hydrocephalus, spina bifida, cleft lip/palate, congenital heart disease and club foot, and counsel families well.

Must know before the ward round

1. Down syndrome: ~1 in 700–1000 live births. Free trisomy 21 (nondisjunction, mostly maternal meiosis I) ~95%; Robertsonian translocation ~3–4% (often t(14;21)); mosaic ~1–2%. Risk rises with maternal age: ~1/1000 at 30, ~1/350 at 35, ~1/100 at 40, ~1/30 at 45.
2. Newborn features: HYPOTONIA (commonest), flat facial profile, upslanting palpebral fissures, epicanthic folds, Brushfield spots, small low-set ears, protruding tongue, flat occiput, excess nuchal skin, single palmar crease (~45%), sandal gap, 5th-finger clinodactyly. Confirm with karyotype (needed for recurrence counselling).
3. CHD in ~40–50%: AVSD (classic, ~40% of the CHD), VSD, ASD, PDA, tetralogy. Echocardiogram for EVERY newborn with DS even if the examination is normal; repair AVSD by 3–6 months before pulmonary hypertension/Eisenmenger.
4. GI: duodenal atresia (double bubble; ~30% of duodenal atresia has DS), Hirschsprung disease, anorectal malformation, oesophageal atresia/TOF. Bilious vomiting in a DS newborn = surgical emergency.
5. Screening schedule (AAP 2022): at birth echo, FBC (TAM, polycythaemia), TSH, hearing, red reflex/cataract; TSH at 6 and 12 months then yearly; hearing 6-monthly then yearly; eyes by 6 months then yearly; Hb yearly from 1 year; sleep study by 4 years; ask about neck symptoms (no routine atlantoaxial X-rays).
6. Haematology: transient abnormal myelopoiesis in ~10% of newborns (resolves by 3 months, but 20–30% later develop AML); leukaemia risk 10–20 × (ALL and megakaryoblastic AML).
7. Recurrence: free trisomy ~1% (or the age risk if higher); translocation carrier mother 10–15%, father ~2–5%; a 21;21 translocation carrier: 100%.
8. Spina bifida (myelomeningocele): cover with sterile saline-soaked gauze and cling film, nurse prone, no latex, measure HC daily, refer for closure within 48–72 h; 80–90% develop hydrocephalus. Prevention: folic acid 400 µg daily from before conception to 12 weeks; 5 mg if previous NTD or on antiepileptics.
9. Hydrocephalus in Uganda is mostly POSTINFECTIOUS (after neonatal sepsis); signs: HC crossing centiles, tense fontanelle, split sutures, sunsetting eyes, scalp veins. Cranial USS; refer to neurosurgery (ETV ± choroid plexus cauterisation, or VP shunt) at CURE Mbale or Mulago.
10. Cleft lip/palate ~1 in 700: feeding first (upright, expressed breast milk by cup/spoon or special bottle), weigh weekly; lip repair ~3 months ('rule of 10s'), palate 9–12 months before speech; Pierre Robin = airway risk, nurse prone.
11. Club foot (CTEV) 1–2/1000, male 2:1, half bilateral; CAVE (Cavus, Adductus, Varus, Equinus); Ponseti: weekly casts × 5–7, Achilles tenotomy (~80–90%), then foot-abduction brace 23 h/day for 3 months and nights until 4–5 years.
12. Pulse oximetry newborn screen: pass if SpO_2 ≥95% in right hand or a foot with ≤3% difference; <90% = fail (critical CHD).

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1. Definition

Congenital anomalies (birth defects) are structural or functional abnormalities present at birth. About **1 in 33 babies** has one, and they are a leading cause of neonatal and infant death and lifelong disability.

Down syndrome (DS) is the condition caused by an extra copy of chromosome 21 (whole or the critical part of its long arm). It is the commonest chromosomal disorder in live-born babies and the commonest genetic cause of intellectual disability. A child with DS is not "sick", but has **predictable medical problems** (heart disease, bowel obstruction, hypothyroidism, hearing and eye problems, sleep apnoea, leukaemia, atlantoaxial instability) that a **screening schedule** detects early.

This topic also covers the other anomalies you must recognise and give first care for: **hydrocephalus, spina bifida (neural tube defects), cleft lip and palate, congenital heart disease (CHD) and club foot**.

Term	Meaning	Example
Malformation	Abnormal formation of a tissue (intrinsic)	Neural tube defect, cleft lip, CHD
Deformation	Normal tissue moulded by mechanical forces	Positional talipes, plagiocephaly (oligohydramnios, breech)
Disruption	Breakdown of normally formed tissue	Amniotic band amputation
Dysplasia	Abnormal organisation of cells in a tissue	Skeletal dysplasia (achondroplasia), ectodermal dysplasia
Sequence	One primary anomaly causing a cascade	**Pierre Robin** (micrognathia → glossoptosis → cleft palate); Potter sequence (renal agenesis → oligohydramnios → lung hypoplasia, flat face, limb deformities)
Syndrome	A recognisable pattern of anomalies with one cause	**Down syndrome**, Turner, fetal alcohol syndrome
Association	Anomalies that occur together more than by chance	**VACTERL** (Vertebral, Anorectal, Cardiac, TracheoEsophageal fistula, Renal, Limb)
Major vs minor	Major: medical, surgical or cosmetic significance (CHD, NTD). Minor: little significance alone (single palmar crease, ear tag), but **≥3 minor anomalies** raise the chance of a major one or a syndrome	

CLINICAL PEARL

The one thing never to forget: **every baby with Down syndrome needs an echocardiogram**, even if you hear no murmur. An atrioventricular septal defect can be silent in the first weeks while pulmonary pressures are high, and must be repaired by about 3–6 months before irreversible pulmonary hypertension (Eisenmenger) develops.

2. Epidemiology

Global

Condition	Frequency
Any congenital anomaly	About **1 in 33** births; more than **240,000 newborns die each year** within 28 days from them (WHO)
Congenital heart disease	About **8 per 1000** births (about 1 in 4 of these critical)
Neural tube defects	**1–2 per 1000** without folic acid fortification (much lower with it)
Cleft lip ± palate	About **1 in 700–1000**
Club foot	**1–2 per 1000**; male:female **2:1**; **half bilateral**
Down syndrome	About **1 in 700–1000** live births; falls where prenatal screening and termination are common; rises with older maternal age

Uganda

- Births with anomalies are **under-recorded**, especially from home births and stillbirths. Hospital-based surveillance (e.g. at Kawempe/Mulago) has found **neural tube defects, hydrocephalus, club foot, cleft lip/palate, abdominal wall defects and CHD** among the commonest anomalies.
- As infection-related deaths fall, congenital anomalies (especially **CHD, neural tube defects and hydrocephalus**) are an increasing share of child deaths and disability.
- Many Ugandan women do not take folic acid before conception, and the first ANC visit is often after 12 weeks, when the neural tube has already closed (**day 28**).
- **Down syndrome in Uganda**: there is **no routine prenatal screening**; diagnosis is made at birth or, often, **months later** when hypotonia, delayed milestones or heart failure are noticed. Mothers are frequently blamed; some families believe the child is cursed, and some children are hidden or abandoned.

> **UGANDA CONTEXT**
> **Services**: the **Uganda Heart Institute** (Mulago) is the national centre for cardiac surgery (long waiting lists; some children go abroad with charity support). **CURE Children's Hospital of Uganda (Mbale)** and **Mulago** neurosurgery treat hydrocephalus and spina bifida. **CoRSU (Kisubi)** and other Smile Train partner hospitals offer free cleft surgery. **Ponseti club foot clinics** exist in many regional referral hospitals (the Uganda Sustainable Clubfoot Care Project introduced the method nationally).

3. Aetiology and Risk Factors

Causes of congenital anomalies

Group	Share	Examples
Chromosomal	~6–10%	Trisomies 21, 18, 13; Turner; deletions (22q11)
Single-gene	~7%	Achondroplasia, many syndromes
Multifactorial (most identifiable)	Large share	NTD, cleft lip/palate, CHD, club foot
Teratogens		Alcohol, **valproate, carbamazepine, phenytoin**, warfarin, isotretinoin, ACE inhibitors, uncontrolled maternal **diabetes**; maternal infections: **rubella, CMV, syphilis, toxoplasmosis, Zika**
Nutritional		**Folate** and iodine deficiency
Unknown	~50%	

Down syndrome

- **Maternal age** is the main risk factor for free trisomy (nondisjunction rises with the age of the oocyte):

Maternal age	Approximate risk of DS at birth
30	1 in 1000
35	**1 in 350**
40	**1 in 100**
45	1 in 30

- **A parent carrying a balanced Robertsonian translocation** (suspect after previous affected pregnancies, recurrent miscarriages or stillbirths): risk independent of age (section 5).
- **Previous child with DS**: recurrence risk depends on the cytogenetic type (section 5).

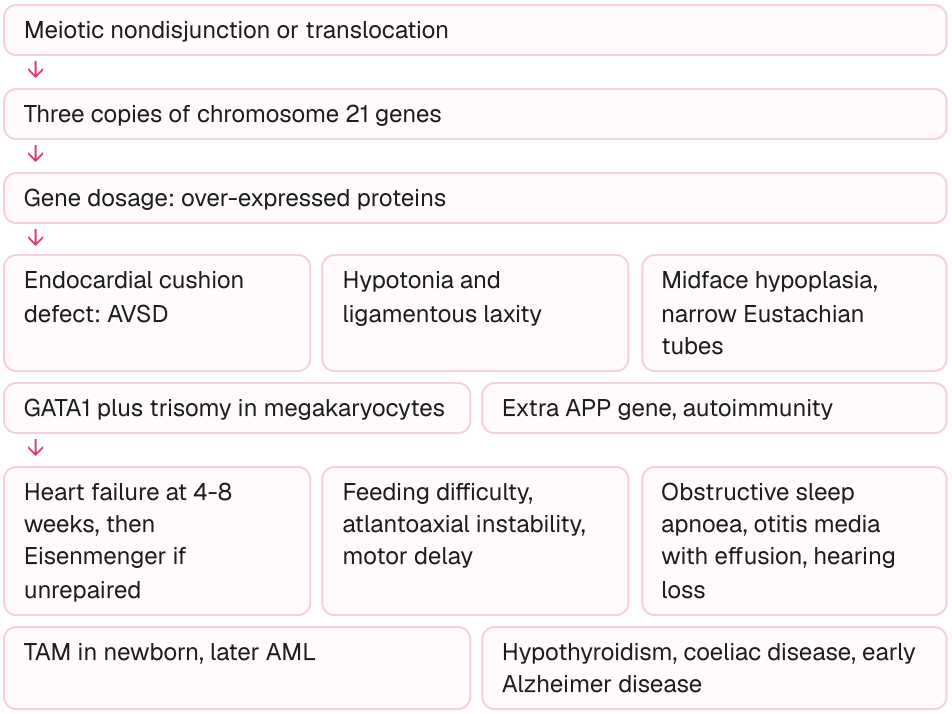
Specific anomalies

- **Neural tube defects**: folate deficiency, **valproate/carbamazepine**, maternal diabetes, obesity, hyperthermia; previous affected child.
- **Cleft lip/palate**: multifactorial; teratogens (phenytoin, valproate, smoking, alcohol); syndromes (Pierre Robin, 22q11 deletion).
- **Club foot**: multifactorial; associations with spina bifida, arthrogryposis, oligohydramnios.
- **Hydrocephalus**: **neonatal sepsis and ventriculitis** (postinfectious, commonest in Uganda), preterm intraventricular haemorrhage, myelomeningocele (Chiari II), aqueduct stenosis, Dandy–Walker, tumour.
- **CHD**: DS and other syndromes, maternal rubella, diabetes, teratogens.
- **Consanguinity** increases autosomal recessive conditions.

4. Pathophysiology

Down syndrome: why the features appear

- **Gene dosage**: three copies of chromosome 21 genes (e.g. *DYRK1A*, *APP*, *COL6A1*, *RCAN1*) over-express proteins. **APP** (amyloid precursor protein) explains **early-onset Alzheimer's disease** (amyloid plaques by 40). Genes affecting **endocardial cushion** development explain **AVSD**. **GATA1** mutations in fetal megakaryocytes combined with trisomy cause **transient abnormal myelopoiesis** and later megakaryoblastic leukaemia.
- **Hypotonia** and ligamentous laxity explain floppy babies, feeding difficulty, **atlantoaxial instability**, flat feet and joint hypermobility.
- **Midface hypoplasia**, a narrow upper airway, a large tongue relative to the mouth, adenotonsillar hypertrophy and hypotonia explain **obstructive sleep apnoea** (50–75%); narrow Eustachian tubes cause **otitis media with effusion** → conductive hearing loss.
- **Autoimmunity** is common: hypothyroidism (Hashimoto), coeliac disease, type 1 diabetes, alopecia.
- **Immune dysfunction** explains more frequent and severe respiratory infections.



Other anomalies: key mechanisms

- **Neural tube defects**: failure of neural tube closure by **day 28** after conception (before many women know they are pregnant). In myelomeningocele the spinal cord is exposed and damaged (motor and sensory loss below the level, neurogenic bladder and bowel), and the **Chiari II malformation** (hindbrain herniation) blocks CSF flow → **hydrocephalus in 80–90%**.
- **Hydrocephalus**: CSF is made by the choroid plexus (~500 mL/day in adults) and absorbed by arachnoid granulations. Obstruction (aqueduct stenosis, Chiari II, Dandy–Walker, tumour) or failed absorption (post-haemorrhagic; **post-infectious**: ventriculitis after neonatal sepsis causes septations and blocked absorption) → ventricular enlargement. In infants the sutures are open, so the **head enlarges** before pressure signs appear.
- **Cleft lip and palate**: failure of fusion of the medial nasal and maxillary processes (lip, **4–7 weeks**) and of the palatal shelves (palate, **8–12 weeks**).
- **Congenital heart disease**: abnormal development of the heart tube, septa and outflow tract at **3–8 weeks**. The **transition at birth** decides when the baby becomes ill: left-to-right shunts (VSD, AVSD, PDA) cause **heart failure at 4–8 weeks** as pulmonary vascular resistance falls; **duct-dependent** lesions (coarctation, hypoplastic left heart, pulmonary atresia, TGA) cause collapse or cyanosis **when the duct closes** in the first days.
- **Club foot**: abnormal fibrosis and shortening of medial and posterior tendons and ligaments → fixed **equinus, varus (heel inversion), adductus (forefoot) and cavus**.

Feature links

- Hypotonia → weak suck, long feeds, poor weight gain, weak Moro, joint hyperflexibility.
- Ligamentous laxity → atlantoaxial instability → neck pain, torticollis, gait change, weakness, bladder change (cord compression).
- Endocardial cushion defect → AVSD → large left-to-right shunt as pulmonary resistance falls → fast breathing, sweating with feeds, hepatomegaly at 4–8 weeks.
- Unrepaired large shunt → pulmonary vascular disease → Eisenmenger (cyanosis, inoperable).
- High pulmonary pressure in the first weeks → small shunt → **no murmur** despite AVSD.
- Narrow Eustachian tubes → otitis media with effusion → conductive hearing loss → speech delay.

- Small airway, large tongue, big adenoids and tonsils, hypotonia → snoring, apnoeas, daytime sleepiness → pulmonary hypertension.
- Autoimmune thyroiditis or congenital hypothyroidism → prolonged jaundice, constipation, poor growth, worse development.
- GATA1 mutation → blasts in the newborn blood film (TAM) → later megakaryoblastic AML.
- Duodenal atresia → bilious vomiting, double bubble, polyhydramnios in pregnancy.
- Aganglionic distal bowel (Hirschsprung) → delayed meconium, distension, constipation.
- Extra APP gene → amyloid plaques → early Alzheimer's disease from 40 years.
- Chiari II in myelomeningocele → blocked CSF → hydrocephalus (rising HC, tense fontanelle).
- Exposed cord in myelomeningocele → paralysis and sensory loss below the level, talipes, neurogenic bladder → UTIs and renal damage.
- Raised intracranial pressure in an infant → split sutures, sunsetting eyes, dilated scalp veins, vomiting.
- Cleft palate → no seal for suction → milk through the nose, aspiration, poor growth; abnormal palatal muscle insertion → OME, nasal speech.
- Duct closure in a duct-dependent lesion → cyanosis or collapse with weak femoral pulses in the first week.

5. Classification / Staging

Down syndrome: cytogenetic types

Type	Share	Mechanism	Recurrence risk
Free trisomy 21 (47,XX/XY,+21)	~95%	**Meiotic nondisjunction**, ~90% maternal (mostly meiosis I); risk rises with maternal age	~1% (or the maternal age risk, whichever is higher)
Robertsonian translocation (46 chromosomes, extra 21q attached to 14, 21 or 22)	~3–4%	De novo (~75%) or inherited from a balanced carrier parent (~25%); **not** related to maternal age	Mother carrier ~10–15%; father carrier ~2–5%; **rob(21;21) carrier: 100%**
Mosaic (some cells 46, some 47)	~1–2%	Post-zygotic nondisjunction	Low; features may be milder

EXAM FAVOURITE

Why karyotype a baby who obviously has Down syndrome? Because the **type** determines the **recurrence risk**: if it is a translocation, both parents need karyotypes, and other relatives may be carriers. A rapid test (FISH/QF-PCR) confirms trisomy fast, but only a full karyotype shows a translocation.

Neural tube defects

Type	Description
Spina bifida occulta	Vertebral arch defect only; skin covered; may have dimple, hair tuft, lipoma, haemangioma; check for tethered cord if neurological signs
Meningocele	Meninges herniate through the defect; cord in place; often normal neurology
Myelomeningocele (MMC)	Cord and meninges in an open sac; paralysis and sensory loss below the level, neurogenic bladder and bowel, talipes, hip dislocation; **hydrocephalus in 80–90%** (Chiari II)
Encephalocele / anencephaly	Cranial defects; anencephaly is incompatible with life

Hydrocephalus by cause: **postinfectious** (commonest in Uganda, after neonatal sepsis/ventriculitis), congenital (aqueduct stenosis, Chiari II with MMC, Dandy–Walker), post-haemorrhagic (preterm IVH), tumour. Functionally: obstructive (non-communicating) versus failed absorption (communicating).

Congenital heart disease

Group	Lesions	Typical presentation
Acyanotic, left-to-right shunt	**VSD** (commonest CHD), ASD, PDA, **AVSD** (DS)	Murmur; **heart failure at 4–8 weeks** (fast breathing, sweating and tiring during feeds, poor weight gain, hepatomegaly); recurrent chest infections; pulmonary hypertension if untreated
Acyanotic, obstructive	Coarctation, aortic stenosis, pulmonary stenosis	Murmur, weak/absent femoral pulses (coarctation), **collapse when the duct closes**
Cyanotic	**Tetralogy of Fallot** (commonest cyanotic beyond the newborn period: VSD, overriding aorta, RV outflow obstruction, RV hypertrophy), **transposition of the great arteries** (commonest cyanotic in the newborn), tricuspid atresia, TAPVD, truncus arteriosus	Cyanosis, **hypercyanotic spells** (TOF), clubbing, polycythaemia, squatting, brain abscess, stroke

Cleft lip and palate: by extent: unilateral or bilateral lip, alveolus, hard and/or soft palate; **submucous cleft** (bifid uvula, notch in the hard palate) is easy to miss.

Club foot: **positional (postural) talipes** (fully correctable) versus **structural CTEV** (rigid). Severity by the **Pirani score** (0–6; higher = more severe), recorded for each foot at every visit.

6. Clinical Features

History

Each question has a reason.

Pregnancy

- **Maternal age** → DS risk (section 3).
- **Parity, ANC attendance and gestation at first visit** → missed screening and folic acid window.
- **Folic acid** (when started) → NTD risk.
- **Drugs**: valproate, carbamazepine, phenytoin, warfarin, ACE inhibitors, herbal medicines; **alcohol, smoking** → teratogens.
- **Diabetes** → CHD, NTD, caudal regression.
- **Febrile illness or rash** (rubella) → cataract, CHD, deafness.
- **HIV and syphilis results** → congenital infection; PMTCT and feeding plan.
- **Ultrasound findings**: polyhydramnios (duodenal or oesophageal atresia), nuchal translucency, double bubble → anticipated anomaly.
- **Fetal movements; breech; oligohydramnios** → talipes, DDH, Potter sequence.

Family

- **Previous babies with anomalies, miscarriages or stillbirths** → balanced translocation or single-gene disorder.
- **Consanguinity** → autosomal recessive conditions.
- **Relatives with DS, clefts or NTDs** → recurrence counselling.

Birth: gestation, birthweight, Apgar, resuscitation → asphyxia as a cause of hypotonia; breech → talipes, DDH.

Neonatal

- **Feeding**: suck, choking, **milk through the nose** → cleft palate; frothing and choking with the first feed → oesophageal atresia.
- **Vomiting and its colour**: **bilious = obstruction** (duodenal atresia, malrotation).
- **Meconium passage within 48 h** → delayed in Hirschsprung, anorectal malformation.
- **Breathing difficulty, cyanosis, sweating with feeds, poor weight gain** → CHD and heart failure.
- **Jaundice** → prolonged in hypothyroidism; early with duodenal atresia.
- **Urine stream** → posterior urethral valves, neurogenic bladder.

Later childhood: milestones, hearing, vision, **snoring and pauses in breathing** (OSA), constipation, weight and growth, school; **neck pain, gait change, weakness, bladder or bowel change** (atlantoaxial instability).

In every child: **HIV status, immunisation and feeding**; **the parents' understanding, beliefs and support** → counselling plan.

Examination

Follow the order of Hutchison's Clinical Methods (24th edition), adapted for the newborn and infant: observe first, examine the heart while the baby is quiet, leave distressing parts (palate, hips, eyes with the ophthalmoscope) to last.

Introduction, consent and preparation

- Introduce yourself; confirm the baby's name, age and the parent's relationship to the child.
- Explain the examination and gain consent from the parent (assent from an older child).
- **Hand hygiene** before touching the baby.
- **Position and exposure**: undress fully on a warm surface under a radiant heater or in a warm room; keep the baby warm (especially with open lesions or abdominal wall defects).

General inspection (end of the cot)

- **Gestalt**: look at the baby as a whole first. Well or unwell; colour (cyanosis, pallor, jaundice, plethora from polycythaemia); breathing pattern and effort; posture and tone (frog-leg posture in hypotonia); obvious anomalies.
- **Vitals**: temperature, RR, HR, **SpO_2 right hand and a foot**, capillary refill, blood glucose if unwell.
- **Growth**: weight, length, **head circumference** plotted with Z-scores (use DS-specific charts or WHO charts for DS).

Newborn with possible Down syndrome

Area	Findings (approximate frequency in newborns with DS)
Tone and reflexes	**Hypotonia** (~80%), **weak or absent Moro** (~85%), joint hyperflexibility (~80%)
Face	**Flat facial profile** (~90%), **upslanting palpebral fissures** (~80%), epicanthic folds, **Brushfield spots** (speckled iris), small nose with flat nasal bridge, small mouth with protruding tongue
Head and neck	Brachycephaly with **flat occiput**, small **dysplastic, low-set ears** (~60%), **excess skin at the nape** (~80%), third fontanelle
Hands and feet	**Single palmar crease** (~45%; also in ~1–2% of normal people), short broad hands, **clinodactyly** of the 5th finger (~60%), **sandal gap** between first and second toes
Other	Dysplastic pelvis on X-ray (~70%), heart murmur (often absent early), abdominal distension or absent anus (GI anomalies)

Hall's criteria: almost all newborns with DS have **6 or more** of the 10 cardinal signs (flat face, poor Moro, hypotonia, hyperflexibility, nuchal skin, upslanting fissures, dysplastic ears, dysplastic pelvis, clinodactyly, single palmar crease).

Hands, face and neck

- **Hands**: palmar creases, clinodactyly, polydactyly (trisomy 13), clenched hands with overlapping fingers (trisomy 18), nails, peripheral cyanosis.
- **Face**: profile, palpebral fissures, epicanthus, Brushfield spots, nasal bridge, tongue, **jaw (micrognathia: Pierre Robin)**, **lip** (cleft).
- **Head**: size and shape, fontanelles (size, tension), **sutures** (split in hydrocephalus), dilated scalp veins, **sunsetting eyes**.
- **Ears**: position, shape, tags, pits.
- **Neck**: nuchal skin, webbing, masses.

Systematic newborn anomaly examination (every baby)

- **Eyes**: **red reflex** (cataract, retinoblastoma).
- **Palate**: **look and feel with a finger**; soft palate clefts are easy to miss.
- **Chest and heart**: work of breathing; **murmur**, **femoral pulses**, **pre- and post-ductal SpO_2**, liver edge (heart failure).
- **Abdomen**: distension, masses, wall defects, umbilical hernia.
- **Genitalia**: hypospadias, undescended testes, ambiguous genitalia.
- **Anus**: patent, position.
- **Spine**: dimple, hair tuft, swelling, open lesion.
- **Limbs**: talipes (correctable or rigid; Pirani score), extra digits, arthrogryposis.
- **Hips (last)**: Ortolani and Barlow.
- **Skin**: haemangioma, cutis aplasia (trisomy 13).

Focused additions by anomaly

- **NTD**: **level of lesion** (motor function: hip flexion L1–2, knee extension L3–4, ankle dorsiflexion L4–5, plantar flexion S1–2), anal tone, bladder dribbling, **HC**, hips, feet.
- **Hydrocephalus**: HC centile and trend, fontanelle, sutures, eyes; in older children headache, vomiting, papilloedema, sixth nerve palsy.
- **Cleft**: extent (lip unilateral/bilateral, alveolus, hard/soft palate) and **watch a feed**.
- **Club foot**: flexibility and **Pirani score**; always examine **spine and hips** too.

Completing the examination

Re-dress and wrap the baby, thank the parents, **hand hygiene**, summarise and document.

Document these positives	And these negatives
Dysmorphic features observed	"Palate intact on palpation"
SpO_2 right hand and foot	"Femoral pulses palpable, no murmur"
HC and centile	"Anus patent, meconium passed"
Level and neurology of NTD	"Spine intact, no sacral dimple"
Pirani score for each foot	"Hips stable (Ortolani/Barlow negative)"

Red flags

RED FLAG

Emergencies in a newborn with an anomaly:

- **Bilious (green) vomiting**, abdominal distension, no meconium by 48 h → bowel obstruction (duodenal atresia, malrotation, Hirschsprung): nil by mouth, NG tube on free drainage, IV fluids, urgent surgical referral
- **Frothing, choking and cyanosis with the first feed**, unable to pass an NG tube → oesophageal atresia: keep nil by mouth, suction the pouch, refer
- **Central cyanosis** not improving with oxygen, or **collapse with weak femoral pulses** in the first week → duct-dependent CHD
- **Breathing difficulty with micrognathia** → Pierre Robin airway obstruction: prone position, nasopharyngeal airway
- **Open lesion on the back or head** leaking CSF → infection risk: sterile saline-soaked dressing, prone, refer
- **Rapidly enlarging head**, tense fontanelle, vomiting, sunsetting eyes → hydrocephalus with raised pressure
- In DS: **neck pain, torticollis, new gait change, weakness or bladder/bowel change** → atlantoaxial instability with cord compression (urgent imaging)
- In a child with a shunt: **vomiting, drowsiness, bulging fontanelle, headache** (blockage) or **fever, redness along the track** (infection) → return immediately

Thinking like a clinician

- **The "floppy baby with a flat face"**: think DS but also check the glucose, temperature and sepsis signs; hypotonia from DS does not explain lethargy, poor perfusion or poor feeding with fever.
- **DS + vomiting**: bilious = duodenal atresia or malrotation; non-bilious, projectile at 3–6 weeks = pyloric stenosis; constipation + distension = **Hirschsprung disease** (DS carries a 40-fold higher risk).
- **DS + fast breathing and poor weight gain at 6 weeks**: heart failure from **AVSD/VSD** is far more likely than pneumonia (though both can coexist).
- **DS + "prolonged jaundice" or constipation**: congenital **hypothyroidism**.
- **DS + high white count with blasts on the film in the first week**: **TAM**; refer to haematology; usually resolves but needs follow-up for leukaemia.
- **DS + snoring and daytime sleepiness**: OSA (pulmonary hypertension risk); refer ENT.
- **DS + "behaviour problems" or school decline**: check hearing, vision, thyroid, sleep and depression before blaming the DS.
- **A baby with an NTD** has a **whole-body problem**: look at the legs (movement, talipes), anus (tone), bladder (dribbling), head (circumference daily), and hips.
- **A "club foot" that corrects fully with gentle pressure** is a **positional** talipes from uterine moulding; it needs only reassurance and stretching. A rigid foot needs Ponseti. Always examine the **spine and hips** too.
- **Cyanosis in a newborn that does not improve with 100% oxygen** (hyperoxia test) suggests cyanotic CHD rather than lung disease.
- **No murmur does not mean no heart disease** in DS: pulmonary pressures are high in the first weeks; get the echo.
- **A young mother does not mean a low recurrence risk**: translocations are not age-related; karyotype every child.
- **One minor feature is not a syndrome** (a single palmar crease or epicanthus alone); look for the pattern, and remember ≥3 minor anomalies should prompt a search for a major one.

- **Common student mistakes**: not feeling the palate, forgetting the anus and spine, measuring HC only once in an MMC, describing a dysmorphic baby as "a Down's baby" in front of the parents.

7. Differential Diagnoses

Differential	Distinguishing clinical feature	Distinguishing investigation
Down syndrome	Hypotonia, flat face, upslanting fissures, single crease, sandal gap, AVSD	Karyotype
Trisomy 18 (Edwards)	IUGR, **clenched hands with overlapping fingers**, **rocker-bottom feet**, prominent occiput, small jaw, CHD; most die in the first year	Karyotype
Trisomy 13 (Patau)	**Midline defects**: holoprosencephaly, cleft lip/palate, scalp defects (cutis aplasia), **polydactyly**, microphthalmia, CHD	Karyotype
Congenital hypothyroidism	Large tongue, umbilical hernia, prolonged jaundice, hypotonia, large fontanelle, constipation; no upslanting eyes or single crease	TSH, T4
Zellweger and other hypotonic syndromes / Prader–Willi	Severe hypotonia, feeding difficulty, distinctive faces	Genetic tests
Sepsis, hypoglycaemia, birth asphyxia (acquired hypotonia)	Lethargy, poor perfusion, fever or hypothermia, abnormal birth history; no dysmorphism	Glucose, FBC, blood culture
Normal baby with minor features (single palmar crease; epicanthus in some ethnic groups)	No other features, normal tone	Clinical; karyotype if doubt
Positional talipes vs structural club foot	Positional: fully correctable, normal creases; structural: rigid, deep creases, small calf	Clinical (Pirani)
Macrocephaly: hydrocephalus vs benign familial macrocephaly vs subdural collections	Hydrocephalus: crossing centiles, tense fontanelle, sunsetting; familial: large parents' heads, normal development	Cranial USS/CT, parents' HC
Cyanosis: cyanotic CHD vs lung disease	CHD: no improvement with 100% oxygen, little respiratory distress	Hyperoxia test, pre/post-ductal SpO_2, echo

8. Clinical Workup

Diagnosis of most anomalies is clinical; tests confirm the type, find associated anomalies and plan surgery. Facility levels below are typical, not an MoH list Verify *availability compiled from the source notes*.

Bedside

Test	How / when	Expected finding	How it changes management	Available at
Pulse oximetry	Right hand (pre-ductal) and either foot (post-ductal) after 24 h (or before discharge)	**Pass**: ≥95% in either with ≤3% difference. **Fail**: <90% in either, or repeated 90–94% / >3% difference	Fail → echo (critical CHD)	HC IV and above; often missing at HC III
Glucose	Heel prick	Hypoglycaemia (sick, LBW babies; Beckwith–Wiedemann)	Treat; feeds	All levels with a glucometer

Test	How / when	Expected finding	How it changes management	Available at
NG tube passage	Pass a size 8–10 tube	Stops at about 10 cm = oesophageal atresia	Nil by mouth, suction, refer	All levels
Hb / haematocrit	Capillary	Polycythaemia (common in DS)	Partial exchange if hyperviscosity	HC IV and above
Hearing screen (OAE)	Before discharge or at 6 weeks	Refer if fail	ABR, ENT	Where available (some hospitals)
Red reflex	Ophthalmoscope in a dim room	Absent/white = cataract (DS, rubella) or retinoblastoma	Urgent ophthalmology	Any level with an ophthalmoscope
Head circumference trend	Daily in NTD/hydrocephalus	Rising faster than **1 cm/week** in a neonate = progressive hydrocephalus	Cranial USS, neurosurgery	All levels
Hyperoxia test	100% oxygen in a cyanosed newborn	No rise in SpO_2 = cyanotic CHD	Echo; careful oxygen	Where oxygen is available

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
FBC and blood film	DS newborn; yearly Hb from 1 year	Polycythaemia, thrombocytopenia, **blasts (TAM)**; later anaemia or leukaemia	Haematology referral; partial exchange if hyperviscosity	HC IV (FBC); film at hospitals
TSH, free T4	DS (newborn, 6 and 12 months, then yearly); prolonged jaundice; constipation; poor growth	Raised TSH	Levothyroxine	RRH, NRH, private
Electrolytes, urea, creatinine	On diuretics or ACE inhibitors; vomiting baby with obstruction	Low K^+, dehydration, raised creatinine	Adjust fluids and drugs	General hospital and above
Ferritin / CRP	DS with risk of iron deficiency	Low ferritin	Iron	RRH, private
Coeliac serology	DS with poor growth, diarrhoea, constipation or anaemia only	Positive	Gluten-free diet	NRH, private

Imaging

Test	When	Expected / abnormal finding	How it changes management	Available at
Abdominal X-ray	Bilious vomiting, distension	**Double bubble** (duodenal atresia), dilated loops, absent rectal gas	Surgical referral (Mulago paediatric surgery)	General hospital and above
Chest X-ray	Heart failure, cyanosis	Cardiomegaly, plethora or oligaemia	Supports CHD; pneumonia	General hospital and above

Test	When	Expected / abnormal finding	How it changes management	Available at
Cranial ultrasound (through the fontanelle) / **CT**	Hydrocephalus, NTD	Ventriculomegaly, septations (postinfectious), Chiari II	Neurosurgery (ETV ± CPC, VP shunt)	USS at most hospitals; CT at RRH/NRH
Renal and bladder ultrasound	NTD (neurogenic bladder), anorectal malformation, VACTERL	Hydronephrosis	Clean intermittent catheterisation, urology	General hospital and above
Hip ultrasound / X-ray	Breech, club foot, NTD, clinical instability	DDH	Pavlik harness, orthopaedics	RRH and above
Cervical spine X-ray (flexion/extension)	DS with **neurological symptoms only**	Atlanto-dens interval increased	Neurosurgical review; **not routine screening**	RRH and above

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management	Available at
Karyotype (FISH/QF-PCR for rapid result)	Suspected DS or other chromosomal syndrome	47,+21; translocation; mosaic	Confirms diagnosis; translocation → parental karyotypes and recurrence counselling	Mulago/MakCHS genetics and private labs; often paid, results in weeks
Echocardiogram	**Every newborn with DS**; murmur, cyanosis, failed pulse oximetry, heart failure	AVSD, VSD, ASD, PDA, TOF	Heart failure treatment; timely surgical referral	UHI, Mulago, Mengo, some RRHs, private clinics
Contrast study / rectal biopsy	Delayed meconium, constipation from birth	Transition zone; aganglionosis	Hirschsprung surgery	NRH paediatric surgery
Hearing (ABR) and ophthalmology	DS, congenital infections, cleft palate	Conductive/sensorineural loss; cataract, refractive error, strabismus	Grommets, hearing aids, glasses, cataract surgery	Mulago ENT, CoRSU, some RRHs
Polysomnography	DS by 4 years (AAP); snoring	OSA	Adenotonsillectomy, CPAP	Rarely available in Uganda; overnight oximetry as a proxy

Putting it together

Model one-line summary

> "A 3-day-old term boy, birthweight 2.9 kg, born to a 39-year-old para 6 with no antenatal ultrasound, with hypotonia, flat facial profile, upslanting palpebral fissures, Brushfield spots, a single palmar crease and sandal gaps, feeding poorly but without vomiting, SpO_2 96% right hand and 95% foot, no murmur: clinically Down syndrome, awaiting karyotype and echocardiogram."

Problem list

1. Clinical DS: confirm (karyotype), counsel parents.
2. Screen for CHD (echo) despite no murmur.
3. Feeding difficulty from hypotonia: breastfeeding support, cup feeding expressed milk.
4. Baseline FBC (TAM, polycythaemia), TSH, hearing, red reflex.

5. Family: older mother, recurrence counselling after karyotype; support and myths.

Model case presentation (ward round)

> "Baby Joseph is a 6-week-old boy, weight 3.6 kg (birthweight 3.0 kg; WAZ −1.7), length 54 cm (LAZ −0.8), HC 37 cm (normal), with clinically diagnosed Down syndrome (karyotype pending), born at term at Mengo to a 38-year-old para 5 who is HIV-negative; BCG, OPV0 and HepB birth dose given, 6-week vaccines due today; exclusively breastfed but tiring during feeds. He presents with 2 weeks of fast breathing, sweating during feeds and poor weight gain. On examination: RR 64, HR 168, SpO_2 94% in air, mild subcostal recession, no fever, liver 3 cm below the costal margin, pansystolic murmur at the left lower sternal border. Typical DS features; anus patent, normal stools. **Impression: heart failure from a probable atrioventricular septal defect in an infant with Down syndrome, with poor weight gain.** Plan: echo, CXR, FBC, TSH, electrolytes; furosemide 1 mg/kg 12-hourly, start enalapril 0.1 mg/kg/day if BP allows, expressed breast milk by cup/NG to maintain intake, vaccines today, urgent referral to the Uganda Heart Institute for surgical review before 3–6 months, and counselling and support for the parents."

EXAM FAVOURITE

Worked example: the double bubble. A 1-day-old with DS vomits green fluid after every feed; the abdomen is not distended; no meconium yet. Plain abdominal X-ray: two gas-filled bubbles (stomach and duodenal cap) and **no distal gas**. **Interpretation: duodenal atresia** (complete obstruction; partial obstruction or malrotation would show some distal gas, and malrotation with volvulus is the emergency not to miss). **Action:** nil by mouth, NG tube on free drainage with hourly aspirates, IV fluids (10% dextrose-containing maintenance plus NG losses replaced with 0.9% saline + KCl), check electrolytes, echo before anaesthesia, refer to paediatric surgery for duodenoduodenostomy.

Worked example 2: the "club foot". A 2-day-old born by breech has an inturned right foot. With gentle pressure the foot comes fully to neutral and beyond, creases are normal, the calf is normal. **Interpretation**: positional talipes from uterine moulding, not CTEV. **Action**: reassure, teach stretching, examine the **hips** (breech: DDH risk) and the **spine**, review in 2–4 weeks. A rigid foot with deep medial and posterior creases would need Ponseti casting within the first weeks.

9. Management

Immediate / emergency

First 5 minutes (any newborn with a significant anomaly)

1. **Airway** with attention to the anomaly: **Pierre Robin** → prone or side-lying, nasopharyngeal airway. **Breathing**: in diaphragmatic hernia, **no bag-mask ventilation**; intubate. **Cyanosis**: SpO_2 pre- and post-ductal. **Circulation**: femoral pulses, CRT; collapse = duct-dependent lesion or sepsis.
2. **Thermal care** (especially with open lesions or abdominal wall defects: cover in cling film to reduce heat and fluid loss); **glucose** check.
3. **Protect open lesions**: myelomeningocele or gastroschisis → sterile saline-soaked gauze, cling film, **prone** (MMC) or bowel supported to the side (gastroschisis), **no latex** gloves.
4. **Nil by mouth and NG tube** for suspected bowel obstruction; **Replogle/suction** of the upper pouch for oesophageal atresia.
5. **Call the senior and plan referral early.**

Hypercyanotic (tet) spell: calm the child on the parent's lap in **knee–chest position**, **oxygen**, **morphine 0.1–0.2 mg/kg** IV/IM/SC, **0.9% saline 10–20 mL/kg**, correct acidosis and anaemia; specialist IV propranolol or phenylephrine if persistent; urgent surgical referral.

Oxygen in suspected CHD: give oxygen only to SpO_2 targets; excess oxygen can worsen shunts or promote duct closure in duct-dependent lesions. **Prostaglandin infusion** to keep the duct open is rarely available outside UHI.

Definitive

Down syndrome: at diagnosis (newborn)

1. **Breaking the news** (see "Counselling the family" below), together with the senior, both parents present.
2. **Confirm**: karyotype (rapid FISH/QF-PCR if available).
3. **Examine for associated anomalies**: heart (echo), gut (vomiting, meconium, anus), eyes (red reflex), hearing, hips.
4. **Feeding support**: breastfeeding is possible and protective; hypotonia and a weak suck mean longer feeds; express breast milk and **cup-feed** if needed; watch weight closely (daily in hospital, weekly after discharge until gaining well). If CHD with heart failure: small frequent feeds, NG feeds.
5. **Baseline tests**: FBC and film, TSH (newborn), hearing screen.
6. **Immunise normally** (BCG, OPV, HepB birth dose, and all UNEPI vaccines); PCV and influenza vaccine are especially valuable.
7. **Plan follow-up** in a paediatric clinic and link to early intervention (physiotherapy, speech therapy) and parent support.

Down syndrome: specific problems

Problem	Management
AVSD / large VSD with heart failure	Diuretics and ACE inhibitor (drug table), high-energy feeds, treat infections, **early surgical referral** (UHI) before 6 months; palivizumab not available; avoid long delays
Duodenal atresia / Hirschsprung / anorectal malformation	Surgical referral (Mulago paediatric surgery)
Congenital or acquired hypothyroidism	**Levothyroxine**, monitor TSH/T4
TAM	Haematology; most resolve; follow FBC to 4–5 years (AML risk)
OME and hearing loss	Grommets, hearing aids, speech therapy
Cataract, strabismus, refractive error	Ophthalmology, glasses, surgery
OSA	Adenotonsillectomy, weight control, CPAP
Atlantoaxial instability with symptoms	Urgent imaging and neurosurgery; neck-safe anaesthesia positioning in all
Constipation	Fluids, fibre, lactulose; exclude hypothyroidism and Hirschsprung
Development	Early intervention (physio, OT, SLT), inclusive education

Drug table

Drug	Dose (per kg + max for children; adult)	Route	Frequency	Duration	Key cautions
Furosemide (heart failure: VSD, AVSD, PDA)	**1 mg/kg/dose** (UCG 2023); increase as needed; **max 8 mg/kg/day** (UCG)	Oral or IV	Once daily or 12-hourly	Until surgical repair; review regularly	Hypokalaemia, dehydration; check electrolytes

Drug	Dose (per kg + max for children; adult)	Route	Frequency	Duration	Key cautions
Spironolactone (add-on, K^+-sparing)	**1.5–3 mg/kg/day** (UCG 2023)	Oral	Divided 12-hourly	With furosemide	**Hyperkalaemia** with ACE inhibitors
Enalapril (afterload reduction in large left-to-right shunts)	Start **0.1 mg/kg/day**, up to 0.5–1 mg/kg/day (UCG: 0.1–1 mg/kg/day)	Oral	1–2 doses/day	Until repair	Hypotension, renal function; caution if SBP low
Captopril (alternative)	**0.1–0.3 mg/kg/dose** (UCG)	Oral	8–12-hourly	Until repair	Test dose; monitor BP
Digoxin (heart failure, selected)	Maintenance **15 µg/kg/day** (UCG)	Oral	Once daily or divided	Specialist decision	Toxicity with hypokalaemia; caution in renal disease
Morphine (hypercyanotic spell)	**0.1–0.2 mg/kg**	IV/IM/SC	Once, may repeat	Acute	With knee–chest position, O_2, **0.9% saline 10–20 mL/kg**; respiratory depression
0.9% sodium chloride (tet spell)	**10–20 mL/kg**	IV	Bolus	Acute	Volume expansion to raise systemic resistance
Propranolol (prevention of tet spells awaiting surgery)	**0.5–1 mg/kg/dose**	Oral	6–8-hourly Verify *frequency against local cardiology protocol*	Until surgery	Hypoglycaemia, bradycardia
Levothyroxine (congenital hypothyroidism)	**10–15 µg/kg/day** (newborn); older children by age/weight	Oral	Once daily	Lifelong (review)	Crush in a little milk; monitor TSH/T4 at 2 and 4 weeks then regularly
Folic acid (NTD prevention, mother)	**400 µg/day** (all women); **5 mg/day** if previous NTD, on antiepileptics, or diabetic	Oral	Daily	From ≥1 month before conception to 12 weeks	Give before conception: the neural tube closes by day 28
Paracetamol (post-operative / tenotomy pain)	**15 mg/kg**	Oral	6-hourly	Short course	Max 60 mg/kg/day

Referral criteria

Level	What it can do	Refer up when
HC II / III	Newborn examination, protect open lesions, start feeding support, immunise	**Any** major anomaly, suspected DS, failed pulse oximetry, bilious vomiting, enlarging head, rigid club foot
HC IV / General Hospital	Stabilise (IV fluids, NG tube, oxygen), USS, X-ray, start diuretics	Surgical anomalies, suspected CHD needing echo, hydrocephalus, MMC
Regional Referral Hospital	Paediatrician, echo (some), Ponseti clinic, TSH, cranial USS	Cardiac surgery, neurosurgery, karyotype, complex surgery
National Referral (Mulago, UHI) / CURE Mbale / CoRSU	Cardiac surgery (UHI), neurosurgery (CURE, Mulago), paediatric surgery, cleft surgery (CoRSU, Smile Train partners), genetics	

- **Urgent (same day)**: bilious vomiting, oesophageal atresia, duct-dependent CHD or collapse, open myelomeningocele (closure within **48–72 h**), Pierre Robin airway obstruction, rapidly enlarging head, DS with neurological signs of cord compression.
- **Early (weeks)**: AVSD/large VSD (surgical review by **2–3 months** so repair happens by 3–6 months), cleft lip/palate team, Ponseti clinic (ideally within 2–3 weeks), hydrocephalus without acute signs.

> UGANDA CONTEXT
> Many children with AVSD in Uganda present late with pulmonary hypertension; **referring early** (in the first 2–3 months) is the single most important action. Echo is available at the **Uganda Heart Institute**, Mulago, Mengo and some RRHs and private clinics; TSH at RRH/NRH and private labs; formal hearing testing at Mulago ENT, CoRSU and some RRHs.

Monitoring and follow-up

Down syndrome health-surveillance schedule (based on AAP 2022 health supervision guidance; adapt to what is available)

Age	Checks
Birth – 1 month	Karyotype; **echocardiogram**; FBC (TAM, polycythaemia); **TSH**; hearing screen (OAE/ABR); **red reflex** (cataract); feeding and weight; check for GI anomalies (vomiting, stools); counselling
1 month – 1 year	**TSH at 6 and 12 months**; hearing every **6 months** (until reliable ear-specific testing, then yearly); **ophthalmology by 6 months**; growth on DS-specific charts (or WHO); development and early intervention; heart surgery timing (AVSD by 3–6 months); monitor for respiratory infections, OSA symptoms
1 – 5 years	**TSH yearly**; **Hb yearly** (with ferritin/CRP if risk of iron deficiency); hearing and eyes yearly; **sleep study (or oximetry) by 4 years**; speech and language therapy; dental care; ask about **neck symptoms** (atlantoaxial instability); coeliac testing only if symptoms (poor growth, diarrhoea, constipation, anaemia)
5 – 12 years	Yearly TSH, Hb, hearing; eyes every 1–2 years; growth and weight (obesity); school progress and inclusive education; behaviour (ADHD, autism are more common)
Adolescence – adult	Puberty, menstrual care, sexuality and safeguarding, contraception; mental health (depression); keratoconus, cataracts; vocational training; transition to adult care; from 40 years watch for **early Alzheimer's disease**

> UGANDA CONTEXT
> Where tests cannot be done, the minimum is: **echo once**, **clinical thyroid assessment and TSH when possible**, **regular weight and length**, **hearing and vision checks by simple tests**, and **ask about neck symptoms**.

Other anomalies: HC daily in NTD and hydrocephalus (weekly after discharge); weight weekly in cleft palate and in heart failure; Pirani score at every Ponseti visit; electrolytes on diuretics and ACE inhibitors; TSH/T4 at 2 and 4 weeks after starting levothyroxine.

Hydrocephalus

1. **Recognise**: HC crossing centiles upward (> +2 SD or rising **>1 cm/week in a neonate, >2 cm/month in an infant**), bulging tense fontanelle, **split sutures**, **sunsetting eyes**, dilated scalp veins, irritability, vomiting, poor feeding, developmental delay; in older children headache, vomiting, papilloedema, sixth nerve palsy.
2. **Classify cause** (section 5).
3. **Investigate**: cranial USS through the fontanelle; CT (septations, cause); screen for ongoing infection (CSF only by the neurosurgical team).

4. **Refer** to neurosurgery: **endoscopic third ventriculostomy (ETV) with choroid plexus cauterisation (CPC)**, developed largely at CURE Children's Hospital of Uganda, avoids lifelong shunt dependence in suitable infants; otherwise **ventriculoperitoneal (VP) shunt**.
5. **Shunt complications** (teach parents): **blockage** (vomiting, drowsiness, bulging fontanelle, headache), **infection** (fever, redness along the track, peritonitis), over-drainage; any of these = return immediately.
6. **Supportive care**: nutrition, physiotherapy, vision and development follow-up, pressure-area care for a large head.

Spina bifida (myelomeningocele)

Immediate management

1. Delivery and resuscitation as usual; **do not lie the baby on the lesion**.
2. Cover the lesion with **sterile gauze soaked in warm sterile saline**, then cling film; keep it moist; nurse **prone or on the side**.
3. **Latex-free** handling (high risk of latex allergy later).
4. Assess the **level** (leg movements, anal tone, bladder dribbling), **HC daily**, hips and feet.
5. Antibiotics if the sac is leaking or infected (per local neonatal protocol).
6. **Refer for surgical closure within 48–72 hours** (CURE Mbale, Mulago neurosurgery), with a referral note and the mother.
7. **Long term**: hydrocephalus surveillance, **clean intermittent catheterisation**, bowel programme, renal ultrasound, physiotherapy and orthoses, pressure-sore prevention, school inclusion.

> **UGANDA CONTEXT**
> UCG 2023 (newborn care) lists **club foot** ("refer for special evaluation and treatment"), **cleft lip/palate** ("help mother breastfeed or teach an alternative method if not possible") and **open tissue on the head, back or abdomen** ("cover with sterile gauze soaked in sterile saline before referral") among the abnormal findings to act on before discharge.

Cleft lip and palate

1. **Examine**: extent (unilateral/bilateral lip, alveolus, hard and/or soft palate; submucous cleft with bifid uvula); look for **Pierre Robin sequence** (small jaw, tongue falls back, airway obstruction) and other anomalies (heart, syndromes).
2. **Feeding first**: babies with **cleft lip only** usually breastfeed well. With **cleft palate**: hold **upright**, express breast milk and feed by **cup and spoon**, special squeezable bottles or long teats; small frequent feeds; burp often (swallowed air); **weigh weekly**; NG feeds if failing.
3. **Airway** in Pierre Robin: prone or side-lying, nasopharyngeal airway, refer.
4. **Refer early** to a cleft team (free surgery via Smile Train/partner hospitals such as CoRSU).
5. **Surgery timing**: lip repair at about **3 months** (classic "rule of 10s": ≥10 weeks old, ≥10 lb (4.5 kg), Hb ≥10 g/dL); palate repair at **9–12 months** (before speech develops); later alveolar bone graft (8–11 years), orthodontics, revision surgery.
6. **Follow-up**: hearing (OME is very common with cleft palate: grommets), speech therapy, dental care, psychosocial support; counsel recurrence risk (**~2–4%** for a sibling if one parent or child affected; higher with more relatives).

Congenital heart disease on the ward

Oxygen only to SpO_2 targets, **diuretics ± ACE inhibitor** for heart failure, **energy-dense feeds** (NG if needed), treat anaemia and infections, **dental care and endocarditis prevention**, immunisation, early echo and **referral to the Uganda Heart Institute**. Tet spells: see Immediate / emergency.

Club foot (congenital talipes equinovarus)

1. **Diagnose**: rigid **CAVE** deformity (Cavus, Adductus, Varus, Equinus) that cannot be corrected passively; deep medial and posterior creases, small calf. Distinguish **positional talipes** (fully correctable). Examine **spine** (spina bifida), **hips** (DDH) and the whole baby (arthrogryposis, syndromes).
2. **Score** with the **Pirani score** (0–6) at each visit.
3. **Ponseti method** (start in the first weeks of life, ideally within 2–3 weeks):
   - **Weekly manipulation and above-knee plaster casts**, correcting **cavus first, then adductus and varus together**, never pronating the foot; usually **5–7 casts**.
   - **Percutaneous Achilles tenotomy** (under local anaesthesia, ~80–90% of feet) to correct residual equinus, then a final cast for 3 weeks.
   - **Foot-abduction brace** (boots and bar, feet at 60–70° external rotation on the affected side): **23 hours a day for 3 months**, then **at night and naps until 4–5 years**.
4. **Brace adherence** is the main determinant of **relapse**; counsel and follow up regularly. Relapses are treated with repeat casting ± tibialis anterior tendon transfer.
5. **Uganda**: Ponseti clinics in regional referral hospitals and CoRSU/Mulago; treatment is often free; late presenters (walking age) can still be treated with casting, though less successfully.

Counselling the family (applies to all anomalies)

1. **Prepare**: confirm facts, involve the senior, choose a private room, have the baby present (wrapped and named), both parents together if possible, a nurse or counsellor, and an interpreter if needed.
2. **Ask what they have noticed and what they understand.**
3. **Name the condition clearly**: "Your baby, Joseph, has a condition called Down syndrome." Use **person-first, positive language** ("a baby with Down syndrome", not "a Down's baby" or old offensive terms).
4. **Explain the cause simply**: an extra chromosome present from conception; **nobody's fault**; not caused by anything the mother did, not witchcraft, not contagious.
5. **Balanced information**: what to expect (slower development, learning disability usually mild to moderate, health checks for heart, thyroid, hearing, eyes), and what is possible (walking, talking, school, relationships, work, a long life).
6. **Practical plan**: feeding, tests (karyotype, echo), follow-up dates, early intervention.
7. **Recurrence risk** once the karyotype is known; offer genetic counselling for future pregnancies (and that prenatal screening exists, mostly privately in Uganda).
8. **Emotional support**: allow silence and grief, address myths with the extended family, link to **parent support groups**, social worker and church/community support; screen the mother for depression; follow up.
9. **Document** what was said, and arrange to meet again soon (parents remember little of the first conversation).

Before discharge (newborn with DS)

Echo (or referral for it), feeding and weight, vomiting and stools (GI anomalies), red reflex, hearing screen, FBC (TAM, polycythaemia), TSH, immunisations, follow-up date, counselling done.

10. Complications

Disease-related

Condition	Main complications
Down syndrome	Heart failure, pulmonary hypertension (Eisenmenger), recurrent pneumonia, OSA, hearing and visual impairment, hypothyroidism, leukaemia, atlantoaxial subluxation, obesity, coeliac disease, diabetes, early Alzheimer's, depression; stigma

Condition	Main complications
Hydrocephalus	Developmental delay, blindness, seizures, death
Myelomeningocele	Paraplegia, neurogenic bladder (UTIs, renal damage), bowel incontinence, pressure sores, hydrocephalus, scoliosis, latex allergy, tethered cord
Cleft lip/palate	Malnutrition and aspiration, OME and hearing loss, speech problems (nasal speech), dental anomalies, psychosocial problems
CHD	Heart failure, failure to thrive, pulmonary hypertension, endocarditis, polycythaemia, stroke, brain abscess (cyanotic), arrhythmia, sudden death
Club foot	Neglected club foot (walking on the dorsum, callosities, disability)

Treatment-related

Treatment	Complication
VP shunt	**Blockage**, **infection**, over-drainage; lifelong shunt dependence (ETV ± CPC avoids this in suitable infants)
Diuretics	Hypokalaemia, dehydration; hyperkalaemia with spironolactone plus ACE inhibitor
ACE inhibitors	Hypotension, renal impairment
Digoxin	Toxicity, worse with hypokalaemia or renal disease
Propranolol	Hypoglycaemia, bradycardia
Excess oxygen in duct-dependent CHD	Duct closure, collapse
Ponseti treatment	**Relapse** with brace non-adherence
Anaesthesia and surgery in DS	Cord injury from neck manipulation if atlantoaxial instability: neck-safe positioning in all
Latex exposure in MMC	Latex allergy

11. Prognosis

- **Down syndrome**: with good care, people with DS now live into their **50s and 60s** in high-income countries; in Uganda, **unrepaired heart disease and infections** still cause many early deaths. Learning disability is usually mild to moderate; most walk, talk and can attend school, work and form relationships. From 40 years early Alzheimer's disease becomes common.
- **AVSD**: good outcome if repaired by 3–6 months; once Eisenmenger develops, surgery is no longer possible.
- **Trisomy 18**: most die in the first year. **Anencephaly** is incompatible with life.
- **Myelomeningocele**: outlook depends on the level of the lesion, hydrocephalus control, bladder care (renal protection) and access to rehabilitation.
- **Hydrocephalus**: development depends on cause, timing of treatment and shunt or ETV complications.
- **Cleft lip/palate**: excellent appearance and speech outcomes with timely repair and speech therapy.
- **Club foot**: Ponseti corrects most feet; relapse is mostly due to brace non-adherence; late presenters do less well.

12. Prevention

- **Folic acid 400 µg daily** from at least 1 month before conception to 12 weeks (**5 mg** if a previous NTD, on valproate/carbamazepine, or with diabetes); **food fortification**; avoid valproate in women of childbearing

potential where possible.

- **Control maternal diabetes** before and during pregnancy; avoid alcohol, smoking and known teratogens; review medicines (including herbal) before conception.
- **Rubella immunity** (measles–rubella vaccination) prevents congenital rubella syndrome; treat syphilis in pregnancy.
- **Early ANC booking** (before 12 weeks) and an anomaly scan; prenatal screening for DS exists, mostly privately in Uganda.
- **Postinfectious hydrocephalus**: clean delivery and prompt treatment of neonatal sepsis.
- **Recurrence counselling** after any affected baby: karyotype the child with DS (and parents if translocation); after a myelomeningocele, recurrence is about **2–5%** after one affected child, so **folic acid 5 mg daily** from before conception to 12 weeks, early ANC and an anomaly scan; cleft recurrence ~2–4% for a sibling.
- **Secondary prevention in DS**: the surveillance schedule (section 9) catches heart, thyroid, hearing, eye, sleep and blood problems early.

13. Special Populations

- **Children with DS undergoing anaesthesia or surgery**: neck-safe positioning in all (atlantoaxial instability); echo before anaesthesia.
- **DS with heart failure**: feeding (NG), immunisation (PCV, influenza) and early infection treatment matter more because respiratory infections are more frequent and severe.
- **Adolescents with DS**: puberty, menstrual care, sexuality and **safeguarding**, contraception, depression, transition to adult care.
- **Preterm babies**: post-haemorrhagic hydrocephalus after IVH; HC surveillance.
- **HIV-exposed infants with anomalies**: support the feeding plan (expressed breast milk by cup in cleft palate) within PMTCT guidance.

14. Key Points

High-yield points

- Every newborn with DS gets an **echo**, even without a murmur.
- Karyotype every child with DS: the **type** decides recurrence risk.
- **Bilious vomiting** in a newborn is a surgical emergency.
- AVSD repair by **3–6 months**: refer in the first 2–3 months.
- DS surveillance: echo, FBC, TSH, hearing and red reflex at birth; TSH at 6 and 12 months then yearly; sleep study by 4 years; ask about neck symptoms.
- MMC: saline-soaked gauze, cling film, prone, no latex, HC daily, closure within 48–72 h.
- Hydrocephalus in Uganda is mostly **postinfectious**.
- Cleft palate: **feeding first**, weigh weekly.
- Club foot: Ponseti from the first weeks; the brace prevents relapse.
- Counsel with person-first language: nobody's fault, not witchcraft.

Numbers & doses

Item	Value
Congenital anomalies	**~1 in 33** births
Down syndrome	**~1 in 700–1000** live births

Item	Value
DS cytogenetics	Trisomy **~95%**, translocation **~3–4%**, mosaic **~1–2%**
Maternal age risk	~1/1000 at 30, **1/350 at 35**, **1/100 at 40**, 1/30 at 45
Recurrence	Free trisomy **~1%**; maternal translocation carrier **10–15%**; paternal **2–5%**; rob(21;21) **100%**
CHD in DS	**40–50%** (AVSD ~40% of these); repair AVSD by **3–6 months**
Duodenal atresia with DS	**~30%** of duodenal atresia cases have DS
TAM	**~10%** of DS newborns; **20–30%** later AML
Leukaemia risk in DS	**10–20 ×**
OSA in DS	**~50–75%**; sleep study by **4 years** (AAP)
TSH in DS	Birth, **6 and 12 months**, then **yearly**
CHD overall	**~8 per 1000** births
Pulse oximetry pass	**≥95%** with **≤3%** hand–foot difference; <90% = fail
NTD prevention	Folic acid **400 µg/day** (5 mg high risk) from pre-conception to **12 weeks**; neural tube closes by **day 28**
MMC closure	Within **48–72 h**; hydrocephalus in **80–90%**
Hydrocephalus HC alarm	Rising >**1 cm/week** (neonate), >**2 cm/month** (infant)
Cleft lip ± palate	**~1 in 700**; lip repair **~3 months**; palate **9–12 months**
Club foot	**1–2/1000**, M:F **2:1**, **50%** bilateral; **5–7** casts; tenotomy **80–90%**; brace **23 h/day × 3 months**, nights to **4–5 years**
Furosemide	**1 mg/kg/dose** daily or 12-hourly; max **8 mg/kg/day** (UCG)
Enalapril	**0.1 → up to 1 mg/kg/day** in 1–2 doses (UCG)
Spironolactone	**1.5–3 mg/kg/day** divided (UCG)
Digoxin maintenance	**15 µg/kg/day** (UCG)
Morphine for tet spell	**0.1–0.2 mg/kg** IV/IM/SC
Levothyroxine (congenital hypothyroidism)	**10–15 µg/kg/day**

Exam traps

- **"No murmur, so no echo needed in DS."** Wrong: echo every DS newborn; AVSD can be silent early.
- **"Mother is 22, so recurrence risk is negligible."** Wrong: if it is a translocation, the risk depends on parental karyotypes, not age; karyotype every child.
- **"Single palmar crease = Down syndrome."** Wrong: 1–2% of normal people have one; diagnose on the overall pattern and karyotype.
- **"Routine neck X-rays for all children with DS."** Wrong: AAP no longer recommends routine screening; ask about symptoms.
- **"Wait for the MMC to be repaired before measuring the head."** Wrong: measure HC daily from birth; hydrocephalus is very common.
- **"Club foot can wait until the child walks."** Wrong: Ponseti works best started in the first weeks.
- **"Bottle-feed every baby with cleft lip."** Wrong: cleft lip alone usually breastfeeds; palate clefts need cup/spoon or special bottles.
- **"High-flow oxygen for every cyanotic newborn."** Careful: in duct-dependent CHD oxygen may promote duct closure; target saturations and get an echo.

Questions seniors ask

Viva questions

1. *What are the cytogenetic causes of DS?* Free trisomy (~95%, nondisjunction), Robertsonian translocation (~3–4%), mosaicism (~1–2%).
2. *What is the commonest heart lesion in DS?* AVSD (also VSD, ASD, PDA, TOF).
3. *What is the recurrence risk after a free trisomy 21?* About 1% (or the age risk if higher).
4. *Name GI anomalies associated with DS.* Duodenal atresia, Hirschsprung, anorectal malformation, oesophageal atresia/TOF; coeliac disease later.
5. *What is TAM?* Transient abnormal myelopoiesis: a self-limiting leukaemoid proliferation in ~10% of DS newborns; 20–30% develop AML later.
6. *How do you manage a myelomeningocele at birth?* Cover with sterile saline-soaked gauze and cling film, prone position, latex-free, assess level and HC, refer for closure within 48–72 hours.
7. *Ponseti method?* Weekly casts (cavus first), Achilles tenotomy, foot-abduction brace 23 h/day for 3 months then nights until 4–5 years.
8. *When are cleft lip and palate repaired?* Lip ~3 months; palate 9–12 months.

On the ward round

- **How would you confirm Down syndrome and why does the type matter?** Karyotype (rapid FISH/QF-PCR first if available); a translocation changes recurrence risk and needs parental karyotypes.
- **What would you check before this baby with DS goes home?** Echo (or referral for it), feeding and weight, vomiting and stools (GI anomalies), red reflex, hearing screen, FBC (TAM, polycythaemia), TSH, immunisations, follow-up date, counselling done.
- **A baby with DS vomits green on day 1. What do you do?** Nil by mouth, NG tube on free drainage, IV fluids, abdominal X-ray (double bubble = duodenal atresia; exclude malrotation with volvulus), electrolytes, urgent surgical referral.
- **Why is early AVSD repair important?** A large shunt causes heart failure and, within months, irreversible pulmonary vascular disease (Eisenmenger), after which surgery is no longer possible.
- **What causes most hydrocephalus in Ugandan infants?** Postinfectious hydrocephalus after neonatal sepsis/ventriculitis; prevention is clean delivery and prompt treatment of neonatal sepsis.
- **How would you counsel a mother whose baby has a myelomeningocele about future pregnancies?** Recurrence risk is about 2–5% after one affected child; take **folic acid 5 mg daily** from before conception to 12 weeks, book ANC early and have an anomaly scan.
- **What makes the Ponseti method succeed or fail?** Correct casting technique and, above all, adherence to the foot-abduction brace; non-adherence is the main cause of relapse.

Mnemonics

- **DS associated problems "HEART CLOGS"** (approximate): **H**ypothyroidism, **E**yes (cataract, refractive), **A**tlantoaxial instability, **R**espiratory infections and OSA, **T**AM/leukaemia; **C**HD (AVSD), **L**earning disability, **O**titis media/hearing, **G**I atresias and Hirschsprung, **S**hort stature.
- **Maternal age risk "35 → 1 in 350; 40 → 1 in 100; 45 → 1 in 30".**
- **Club foot "CAVE"**: **C**avus, **A**dductus, **V**arus, **E**quinus (corrected in that order: cavus first).
- **Cleft "rule of 10s"**: 10 weeks, 10 lb, Hb 10 g/dL for lip repair.
- **VACTERL**: Vertebral, Anorectal, Cardiac, TracheoEsophageal, Renal, Limb.
- **Trisomies**: **13** = midline ("13 is unlucky, splits the middle": cleft, holoprosencephaly, polydactyly); **18** = "Edwards clenches his fists and rocks on his feet" (overlapping fingers, rocker-bottom feet).

OSCE tips

- **Dysmorphology station**: describe systematically from head to toe ("I would like to examine the whole baby"), list features with their names (upslanting palpebral fissures, epicanthic folds, Brushfield spots), then state "These features suggest Down syndrome; I would confirm with a karyotype and screen for associated conditions: echo, FBC, TSH, hearing and eyes."
- **Breaking-news station**: introduce yourself, check identity, ask what they know, warning shot ("I have some important news about Joseph"), name it clearly, pause, respond to emotion, correct myths, give a plan, check understanding, offer a follow-up meeting and support.
- **Newborn examination station**: say aloud that you feel the palate, check the anus and spine, do pre- and post-ductal SpO_2 and leave the hips to last.

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SELF-TEST

Flashcards

29 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Birth prevalence of Down syndrome?	About **1 in 700–1000** live births; risk rises with maternal age.
2	Three cytogenetic types of Down syndrome and their frequencies?	**Free trisomy 21** (nondisjunction) ~95%; **Robertsonian translocation** ~3–4% (often t(14;21)); **mosaicism** ~1–2%.
3	Maternal age-related risk of DS at birth?	~1/1000 at 30; **~1/350 at 35**; **~1/100 at 40**; ~1/30 at 45.
4	Recurrence risks in Down syndrome?	Free trisomy **~1%** (or age risk if higher); translocation carrier **mother 10–15%**, **father 2–5%**; **rob(21;21) carrier 100%**.
5	Commonest neonatal finding in Down syndrome?	**Hypotonia** (with a weak Moro reflex and joint hyperflexibility).
6	List 8 dysmorphic features of DS.	Flat facial profile, **upslanting palpebral fissures**, epicanthic folds, **Brushfield spots**, small low-set ears, protruding tongue, flat occiput, excess nuchal skin, **single palmar crease**, 5th-finger clinodactyly, **sandal gap**.
7	Proportion of DS babies with CHD and the classic lesion?	**40–50%**; classic lesion **AVSD** (also VSD, ASD, PDA, tetralogy of Fallot).
8	Why echo every newborn with DS, and when should AVSD be repaired?	AVSD can be **silent** while pulmonary pressures are high; repair by **3–6 months** before **Eisenmenger** physiology.
9	GI anomalies associated with DS?	**Duodenal atresia** (double bubble), **Hirschsprung disease**, anorectal malformation, oesophageal atresia/TOF; later **coeliac disease**.
10	What is TAM in Down syndrome?	**Transient abnormal myelopoiesis**: blasts in ~10% of DS newborns (GATA1 mutation); usually resolves by 3 months, but **20–30% develop AML** (megakaryoblastic) by 4–5 years.
11	Thyroid screening in DS (AAP 2022)?	**TSH at birth, 6 and 12 months, then yearly** (congenital and autoimmune hypothyroidism).
12	Hearing and eye surveillance in DS?	Hearing: newborn screen, then **every 6 months** until reliable testing, then yearly. Eyes: red reflex at birth, **ophthalmology by 6 months**, then regularly (cataract, refractive error, strabismus, keratoconus).
13	Sleep study recommendation in DS?	**Polysomnography by 4 years** for all (AAP 2022): OSA affects ~50–75%.
14	Atlantoaxial instability in DS: screen how?	**No routine X-rays** for asymptomatic children (AAP 2022); ask at every visit about **neck pain, torticollis, gait change, weakness, bladder/bowel change**; image urgently if present.
15	Why karyotype a baby with obvious DS features?	To find a **translocation**, which changes **recurrence risk** and requires **parental karyotypes**.
16	When does the neural tube close, and NTD prevention?	By **day 28** after conception. **Folic acid 400 µg/day** from ≥1 month before conception to 12 weeks; **5 mg/day** after a previous NTD, on antiepileptics, or with diabetes.
17	Immediate care of a newborn with an open myelomeningocele?	Cover with **sterile saline-soaked gauze + cling film**, nurse **prone**, **latex-free**, assess level and anal tone, **HC daily**, refer for closure within **48–72 h**.
18	What proportion of children with myelomeningocele develop hydrocephalus, and why?	**80–90%**, due to the **Chiari II malformation** obstructing CSF flow.

#	Question	Answer
19	Commonest cause of infant hydrocephalus in Uganda?	**Postinfectious hydrocephalus** after neonatal sepsis/ventriculitis (treated at CURE Children's Hospital Mbale with ETV ± choroid plexus cauterisation or VP shunts).
20	Signs of hydrocephalus in an infant?	HC **crossing centiles**, tense/bulging fontanelle, **split sutures**, **sunsetting eyes**, dilated scalp veins, irritability, vomiting, delay.
21	VP shunt complications parents must recognise?	**Blockage** (vomiting, drowsiness, headache, bulging fontanelle), **infection** (fever, red shunt track, abdominal pain), over-drainage.
22	Feeding a baby with cleft palate?	**Upright**, expressed breast milk by **cup/spoon** or special squeezable bottle, small frequent feeds, burp often, **weigh weekly**; NG if failing. Cleft lip alone usually breastfeeds.
23	Timing of cleft repair and the 'rule of 10s'?	Lip at **~3 months** (≥10 weeks, ≥10 lb, Hb ≥10 g/dL); palate at **9–12 months**, before speech.
24	Pierre Robin sequence?	**Micrognathia → glossoptosis → U-shaped cleft palate** with **airway obstruction**; nurse **prone**, nasopharyngeal airway, refer.
25	Components of club foot and the order of Ponseti correction?	**CAVE**: Cavus, Adductus, Varus, Equinus; correct **cavus first**, then adductus and varus together, then equinus (tenotomy).
26	Ponseti bracing regimen?	Foot-abduction brace **23 h/day for 3 months**, then **nights and naps until 4–5 years**; non-adherence is the main cause of relapse.
27	Newborn pulse oximetry screen: pass and fail criteria?	**Pass**: ≥95% in right hand or foot with **≤3%** difference. **Fail**: <90% in either; repeated 90–94% or >3% difference → echo.
28	Immediate management of a hypercyanotic (tet) spell?	Calm child, **knee-chest position**, **oxygen**, **morphine 0.1–0.2 mg/kg** IV/IM/SC, **0.9% saline 10–20 mL/kg**; specialist propranolol/phenylephrine if persistent.
29	Furosemide dose in a child with heart failure (UCG 2023)?	**1 mg/kg** oral/IV once daily or 12-hourly, increased as needed; **max 8 mg/kg/day**.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A term baby born to a 37-year-old para 5 at a HC IV has hypotonia, upslanting palpebral fissures, a flat facial profile, a single palmar crease and a sandal gap. He is feeding slowly, has no murmur and his SpO_2 is 97% in the right hand and foot. What is the MOST important cardiac step?

A No cardiac assessment needed because there is no murmur
B Repeat auscultation at the 6-week visit and order echo only if a murmur appears
C Chest X-ray only
D Start furosemide prophylactically
E Arrange an echocardiogram in the newborn period regardless of examination

2. A 1-day-old baby with features of Down syndrome has vomited green fluid after each feed. The abdomen is soft and not distended; no meconium yet. What is the MOST likely diagnosis?

A Hypertrophic pyloric stenosis
B Gastro-oesophageal reflux
C Duodenal atresia
D Hirschsprung disease
E Necrotising enterocolitis

3. A 24-year-old woman's first baby has Down syndrome. The karyotype shows 46,XX,der(14;21),+21. What is the MOST appropriate next step for recurrence counselling?

A Karyotype both parents
B Tell her the recurrence risk is about 1% because she is young
C Reassure her that translocations are never inherited
D Advise sterilisation
E No further action is needed

4. A 7-week-old boy with Down syndrome (3.5 kg) has fast breathing (RR 66), sweating during feeds, poor weight gain and a liver edge 3 cm below the costal margin. SpO_2 is 94%; temperature normal. What is the MOST appropriate initial management?

A IV ceftriaxone for pneumonia and discharge when afebrile
B High-flow oxygen to keep SpO_2 at 100%
C IV fluids at 150 mL/kg/day
D Furosemide 3.5 mg (1 mg/kg) 12-hourly, feeding support with expressed breast milk, echo and early referral to the Uganda Heart Institute
E Morphine 0.1 mg/kg and knee-chest position

5. On day 4 of life, an FBC from a baby with Down syndrome shows a white cell count of 48×10^9/L with 30% blasts; the baby is well, with a mildly enlarged liver. What is the MOST likely diagnosis? Advanced

A Neonatal sepsis
B Transient abnormal myelopoiesis
C Acute lymphoblastic leukaemia requiring immediate chemotherapy
D Polycythaemia
E Laboratory error: repeat in 6 months

6. A baby is born at a HC IV with an open, leaking lesion over the lumbosacral spine. The legs move at the hips but not the ankles. What is the MOST appropriate immediate care before transfer?

A Apply dry gauze and nurse supine to keep the lesion clean

B Apply povidone-iodine and tight bandaging to stop the leak

C Cover with sterile saline-soaked gauze and cling film, nurse prone, use latex-free gloves, measure head circumference and refer for closure within 48–72 hours

D Delay referral until the mother is ready to travel in 1 week

E Perform an LP to check for meningitis

7. A 4-month-old girl treated for neonatal sepsis at 2 weeks has a head circumference that has risen from the 50th centile to above the 99th centile. The fontanelle is tense, the sutures are split and the eyes deviate downward. What is the MOST appropriate next step?

A Reassure: large heads run in families

B Lumbar puncture to relieve pressure

C Start oral acetazolamide and review in 3 months

D Give vitamin D for rickets

E Cranial ultrasound through the fontanelle and urgent neurosurgical referral

8. A 2-day-old has a complete unilateral cleft lip and cleft palate. He is choking when breastfeeding and has lost 6% of his birthweight. His jaw is normal-sized and breathing is normal. What is the MOST appropriate feeding plan?

A Express breast milk and feed upright by cup/spoon (or special squeezable bottle), small frequent feeds, burp often, and weigh regularly

B Stop breast milk and use formula through a standard bottle

C Keep nil by mouth until palate repair

D Give feeds lying flat to reduce choking

E Insert a gastrostomy tube

9. A 2-week-old boy has a right foot that points down and in. It cannot be passively brought to neutral; there is a deep medial crease and the calf is thin. The spine and hips are normal. What is the BEST management?

A Reassure: it will correct itself by walking age

B Immediate open surgical release

C Physiotherapy stretching only

D Ponseti method: weekly manipulation and above-knee casts, likely Achilles tenotomy, then a foot-abduction brace

E Start casting at 1 year of age

10. A 10-year-old girl with Down syndrome has had neck pain, a head tilt and a change in her gait over 3 weeks; she now has brisk reflexes and upgoing plantars and has wet herself twice. What is the MOST appropriate next step? Advanced

A Reassure: DS children are often clumsy

B Immobilise the neck, arrange urgent cervical spine imaging (flexion/extension X-rays or MRI) and refer to neurosurgery

C Start physiotherapy for neck stretching

D Treat for urinary tract infection only

E Arrange a routine sleep study

11. A 2-year-old with tetralogy of Fallot (11 kg) becomes deeply cyanosed and floppy after crying during a blood test. SpO_2 is 55%. He is on his mother's lap. What is the MOST appropriate immediate management? Advanced

- A Give IV furosemide 11 mg
- B Lie him flat and start IV fluids at 150 mL/kg/day
- C Give IV digoxin
- D Knee-chest position, oxygen, morphine 1.1–2.2 mg IV/IM/SC, and 0.9% saline 110–220 mL
- E Immediate cardioversion

12. A 30-year-old woman who had a baby with anencephaly asks how to prevent this in her next pregnancy. She is not on any medicines. What is the BEST advice?

- A Folic acid 5 mg daily starting at least 1 month before conception and continuing to 12 weeks of pregnancy
- B Folic acid 400 µg daily starting at the first ANC visit
- C Iron and folic acid tablets from 20 weeks
- D No supplement is needed; anencephaly does not recur
- E Vitamin A 200,000 IU monthly

13. A 3-year-old with Down syndrome is reported by his teacher to be 'not listening and slow'. He snores loudly with pauses in breathing at night and is sleepy in the daytime. Which combination of assessments is MOST appropriate first?

- A EEG and brain MRI
- B Karyotype repeat and chromosomal microarray
- C Coeliac serology only
- D Psychology referral only
- E Hearing assessment (with otoscopy/tympanometry), TSH, and a sleep study or overnight oximetry

14. Which newborn finding is MOST suggestive of trisomy 18 (Edwards syndrome) rather than trisomy 21?

- A Upslanting palpebral fissures and Brushfield spots
- B Sandal gap between the first and second toes
- C Clenched hands with overlapping fingers and rocker-bottom feet
- D Generalised hypotonia
- E Single palmar crease

15. A 3-day-old baby has SpO_2 92% in the right hand and 84% in the foot on routine pulse oximetry. He is feeding and has no respiratory distress. What is the MOST appropriate next step? Advanced

- A Pass the screen and discharge
- B Urgent echocardiogram and senior/cardiology review for possible critical congenital heart disease
- C Repeat the test in 6 weeks
- D Start antibiotics for neonatal sepsis and discharge
- E Give 100% oxygen and discharge when saturations improve

Answers & explanations

> **1. E** · Arrange an echocardiogram in the newborn period regardless of examination
>
> E is right: **every newborn with DS needs an echocardiogram** because 40–50% have CHD and an **AVSD can be silent** early when pulmonary pressures are high. A and B are wrong: absence of a murmur does not exclude AVSD, and delaying risks missing the window for repair before pulmonary hypertension. C is wrong: a normal chest X-ray does not exclude significant CHD. D is wrong: diuretics are for established heart failure, not prophylaxis.

2. C · Duodenal atresia

C is right: **bilious vomiting on day 1** with a **non-distended** abdomen in a baby with DS suggests **duodenal atresia** (obstruction distal to the ampulla); the X-ray shows a double bubble. A is wrong: pyloric stenosis presents at 2–8 weeks with non-bilious projectile vomiting. B is wrong: reflux is not bilious and does not cause obstruction. D is wrong: Hirschsprung disease causes distension and delayed meconium; vomiting comes later. E is wrong: NEC typically affects preterm babies after feeds are established, with distension and bloody stools.

3. A · Karyotype both parents

A is right: this is a **Robertsonian translocation**; about a quarter are inherited from a balanced carrier parent, so **both parents need karyotypes** (maternal carrier risk 10–15%, paternal 2–5%). B is wrong: the 1% figure applies to free trisomy, and maternal age is irrelevant for translocations. C is wrong: translocations can be inherited. D is wrong: this is not indicated or appropriate advice; give information and options. E is wrong: recurrence counselling depends on the parental results.

4. D · Furosemide 3.5 mg (1 mg/kg) 12-hourly, feeding support with expressed breast milk, echo and early referral to the Uganda Heart Institute

D is right: fast breathing, sweating with feeds, poor weight gain and **hepatomegaly** at 4–8 weeks in DS = **heart failure** from a left-to-right shunt (likely AVSD); give **furosemide 1 mg/kg** (UCG), support feeding, get an echo and refer early for surgery. A is wrong: there is no fever, and treating as pneumonia misses heart failure. B is wrong: excess oxygen is unnecessary (SpO_2 94%) and may increase pulmonary blood flow. C is wrong: high-volume IV fluids worsen heart failure. E is wrong: morphine and knee-chest position treat tet spells in cyanotic disease, not heart failure.

5. B · Transient abnormal myelopoiesis

B is right: blasts and hepatomegaly in a well DS newborn are typical of **TAM** (~10% of DS newborns), which usually resolves by 3 months but needs haematology follow-up because 20–30% develop AML later. A is wrong: sepsis does not produce 30% blasts in a well baby. C is wrong: TAM is usually self-limiting; chemotherapy is reserved for severe cases. D is wrong: polycythaemia is a raised haematocrit, not blasts. E is wrong: the result needs prompt repeat and specialist review, not a 6-month wait.

6. C · Cover with sterile saline-soaked gauze and cling film, nurse prone, use latex-free gloves, measure head circumference and refer for closure within 48–72 hours

C is right: an open **myelomeningocele** needs a **moist sterile dressing**, prone positioning, latex precautions, baseline HC (hydrocephalus in 80–90%) and **closure within 48–72 h** (CURE Mbale, Mulago). A is wrong: dry gauze sticks and damages neural tissue, and lying supine puts pressure on the lesion. B is wrong: antiseptics and tight dressings damage exposed neural tissue. D is wrong: delay increases infection (meningitis/ventriculitis). E is wrong: an LP is not indicated and is unsafe near the lesion.

7. E · Cranial ultrasound through the fontanelle and urgent neurosurgical referral

E is right: rapidly crossing HC centiles with raised-pressure signs after neonatal sepsis = **postinfectious hydrocephalus** (the commonest type in Ugandan infants); confirm with **cranial USS** and refer urgently for ETV ± CPC or a VP shunt. A is wrong: familial macrocephaly follows a high centile from birth without symptoms. B is wrong: LP is dangerous in obstructive hydrocephalus and does not treat it. C is wrong: acetazolamide is not definitive treatment and delay causes brain damage and blindness. D is wrong: rickets does not cause raised ICP.

8. A · Express breast milk and feed upright by cup/spoon (or special squeezable bottle), small frequent feeds, burp often, and weigh regularly

A is right: babies with **cleft palate** cannot create suction; **upright** feeding of **expressed breast milk** by cup/spoon or a special bottle, with frequent burping and weight checks, is the standard first step (UCG: teach an alternative method). B is wrong: breast milk should continue (it also protects against otitis media), and a standard teat often fails. C is wrong: palate repair is at 9–12 months; starvation is not an option. D is wrong: lying flat increases nasal regurgitation and aspiration. E is wrong: gastrostomy is not needed when simple methods work.

9. D · Ponseti method: weekly manipulation and above-knee casts, likely Achilles tenotomy, then a foot-abduction brace

D is right: a **rigid** CAVE deformity is structural **club foot**; the **Ponseti method** started in the first weeks (weekly casts, tenotomy in ~80–90%, brace 23 h/day for 3 months then nights to 4–5 years) succeeds in most. A is wrong: structural club foot does not self-correct; reassurance is only for flexible positional talipes. B is wrong: extensive surgery is no longer first-line and causes stiff feet. C is wrong: stretching alone does not correct a rigid deformity. E is wrong: treatment works best when started early.

10. B · Immobilise the neck, arrange urgent cervical spine imaging (flexion/extension X-rays or MRI) and refer to neurosurgery

B is right: neck pain, torticollis, gait change, UMN signs and new incontinence suggest **atlantoaxial instability with cord compression**; immobilise and image urgently, then neurosurgery. A is wrong: new neurological signs are never just clumsiness. C is wrong: neck manipulation could worsen cord compression. D is wrong: a UTI does not explain UMN signs and torticollis. E is wrong: a sleep study is for OSA, not cord compression.

11. D · Knee-chest position, oxygen, morphine 1.1–2.2 mg IV/IM/SC, and 0.9% saline 110–220 mL

D is right: this is a **hypercyanotic (tet) spell**: **knee-chest** position (increases systemic resistance), **oxygen**, **morphine 0.1–0.2 mg/kg** (reduces agitation and infundibular spasm) and a **saline bolus 10–20 mL/kg**. A is wrong: diuretics reduce preload and worsen the spell. B is wrong: lying flat lowers systemic resistance, and maintenance fluids are not a bolus. C is wrong: digoxin increases contractility and can worsen outflow obstruction. E is wrong: this is not an arrhythmia.

12. A · Folic acid 5 mg daily starting at least 1 month before conception and continuing to 12 weeks of pregnancy

A is right: after a previous NTD the recommended dose is **5 mg daily**, started **before conception** (the neural tube closes by day 28) and continued to 12 weeks. B is wrong: 400 µg is the dose for low-risk women, and starting at the first ANC visit (often after 12 weeks in Uganda) is too late. C is wrong: starting at 20 weeks is far too late. D is wrong: recurrence risk is about 2–5% after one affected pregnancy. E is wrong: high-dose vitamin A is teratogenic and does not prevent NTDs.

13. E · Hearing assessment (with otoscopy/tympanometry), TSH, and a sleep study or overnight oximetry

E is right: in DS, apparent inattention and slow progress are often due to **conductive hearing loss** (OME), **hypothyroidism** or **obstructive sleep apnoea** (snoring, pauses, daytime sleepiness); all are treatable. A is wrong: there is no history of seizures or regression. B is wrong: the diagnosis is already known. C is wrong: coeliac testing is for GI symptoms or poor growth. D is wrong: psychology alone misses treatable medical causes.

14. C · Clenched hands with overlapping fingers and rocker-bottom feet

C is right: **clenched hands with overlapping fingers** and **rocker-bottom feet** (with IUGR and a small jaw) are typical of **trisomy 18**. A, B and D are wrong: upslanting fissures, Brushfield spots, sandal gap and hypotonia are typical of **trisomy 21** (trisomy 18 babies are often hypertonic). E is wrong: a single palmar crease occurs in several syndromes and in normal people, so it does not discriminate.

15. B · Urgent echocardiogram and senior/cardiology review for possible critical congenital heart disease

B is right: SpO_2 **<90%** in the foot (and an 8% hand–foot difference) **fails** the screen, suggesting **critical CHD** with right-to-left ductal shunting (e.g. coarctation, interrupted arch, pulmonary hypertension); urgent echo and review are needed. A is wrong: the pass criteria are ≥95% with ≤3% difference. C is wrong: the duct may close within days and the baby could collapse. D is wrong: sepsis should be considered, but discharge without cardiac evaluation is unsafe. E is wrong: oxygen may mask the problem, and in duct-dependent lesions can promote duct closure.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Breaking the news on the postnatal ward

At Mengo Hospital, Mrs Namutebi (39, para 6) delivered a 2.9 kg baby boy, Joseph, 18 hours ago by SVD. The midwife says 'the baby looks different and is very floppy'. Mrs Namutebi had two ANC visits, no ultrasound, and is HIV-negative.

1. What will you look for on examination?

> General: tone (hypotonia), Moro, feeding. Face: flat profile, upslanting palpebral fissures, epicanthic folds, Brushfield spots, small low-set ears, protruding tongue, flat occiput, nuchal skin. Hands/feet: single palmar crease, clinodactyly, sandal gap. Systemic: murmur, femoral pulses, **SpO_2 hand and foot**, abdomen, **anus**, red reflex, hips, spine. Check glucose and temperature.

2. He has typical features of DS, no murmur, SpO_2 97%/96%, anus patent and has passed meconium. What investigations will you arrange?

> **Karyotype** (rapid FISH/QF-PCR if available), **echocardiogram** (all DS newborns), **FBC and film** (TAM, polycythaemia), **TSH**, hearing screen, red reflex; observe feeding and vomiting.

3. How will you tell the parents?

> With the senior, in a private room, both parents present, baby with them and called by name. Ask what they have noticed. Warning shot, then name it: 'Joseph has a condition called Down syndrome.' Explain it is due to an extra chromosome present from conception, **nobody's fault**, not witchcraft, not contagious. Give balanced information (learning disability, health checks, children walk, talk, attend school, can live long lives). Allow emotion and questions; plan next steps; offer to meet again; document.

4. What feeding advice will you give?

> Breastfeeding is encouraged; hypotonia makes feeds slow, so feed frequently, support the baby's head and chin, express breast milk and **cup-feed** top-ups if needed; monitor weight daily in hospital and weekly after discharge. Watch for vomiting (bilious = emergency) and constipation.

5. The karyotype shows free trisomy 21. What is the recurrence risk and what follow-up schedule will you set?

> Recurrence about **1%** or her age-related risk (~1/100 at 39–40), whichever is higher; offer genetic counselling and early ANC in any future pregnancy. Follow-up: echo result and cardiac plan; **TSH at 6 and 12 months then yearly**; hearing every 6 months; ophthalmology by 6 months; growth and development; immunisations; early intervention; Hb yearly from 1 year; sleep assessment by 4 years; neck-symptom questions at every visit.

6. The father says his relatives think the baby is a curse and wants to leave. What do you do?

> Acknowledge his feelings; correct myths with clear facts; involve the social worker, chaplain or counsellor; offer a family meeting with relatives; link the family to a **parent support group**; screen the mother for depression; arrange early follow-up. Safeguard the baby if there are concerns about abandonment or neglect.

Take-away: Diagnose DS clinically, confirm by karyotype, screen systematically (echo first), and break the news with compassion, accuracy and a plan.

Case 2: The open back at a HC IV

A midwife at Kasangati HC IV calls you about a newborn girl delivered 30 minutes ago with a 4 cm open, red, leaking sac over the lower back. The mother did not take folic acid and booked ANC at 20 weeks.

1. What is the likely diagnosis and what immediate care do you instruct?

> **Myelomeningocele** (open NTD). Keep warm; cover with **sterile gauze soaked in warm sterile saline** and cling film; nurse **prone** or on her side; **latex-free** gloves; nothing on the lesion; breastfeed (or expressed milk) if well; start antibiotics if leaking per neonatal protocol; prepare for urgent referral.

2. What will you examine and document?

> Size and level of the lesion, CSF leak; **leg movements** (hip flexion L1–2, knee extension L3–4, ankle movements L4–S2), talipes, hip stability; **anal tone** and wink; bladder dribbling; **head circumference**, fontanelle, sutures; other anomalies (heart, kidneys).

3. Where and how fast should she be referred?

> To neurosurgery (**CURE Children's Hospital Mbale** or **Mulago**) for closure within **48–72 hours**, with a referral note, mother accompanying, warm and protected lesion, feeding plan. Phone ahead.

4. After closure, her HC rises 1.5 cm in a week and the fontanelle is tense. Why, and what next?

> **Hydrocephalus** from the **Chiari II** malformation (80–90% of MMC). Cranial USS; neurosurgical treatment with **ETV ± CPC** or a **VP shunt**; teach the parents shunt warning signs.

5. What long-term problems should the family be prepared for?

> Leg weakness and reduced sensation (pressure sores, fractures), club feet/hip problems, **neurogenic bladder** (clean intermittent catheterisation, UTIs, renal ultrasound), bowel management, hydrocephalus and shunt problems, latex allergy, scoliosis, learning difficulties; need for physiotherapy, orthoses, wheelchair, school inclusion.

6. How should the mother prevent recurrence?

> **Folic acid 5 mg daily** from at least 1 month before conception to 12 weeks; early ANC booking and an anomaly scan; family planning to allow planned pregnancy. Recurrence risk is about 2–5%.

Take-away: Protect the open lesion, refer for closure within 48–72 hours, watch the head for hydrocephalus, and prevent recurrence with high-dose pre-conception folic acid.

Case 3: The baby with a cleft and a club foot

At the 6-week immunisation clinic at a regional referral hospital, you see baby Okot (3.4 kg; birthweight 3.3 kg) who has a left complete cleft lip and palate and a rigid right foot turned down and in. His mother says feeding is 'a struggle' and that relatives told her not to bring him out of the house.

1. What concerns you most today and why?

> **Poor weight gain**: he has gained only 100 g in 6 weeks (expected ~1 kg). With cleft palate, ineffective suction leads to malnutrition and aspiration; this must be fixed before anything else.

2. How will you help with feeding?

> Observe a feed; teach **upright** feeding of **expressed breast milk** by **cup/spoon** or a special squeezable bottle, small frequent feeds, burping; check for aspiration (cough, chest signs); weigh weekly; consider NG feeds or admission if not gaining; check for other anomalies (heart murmur, SpO_2).

3. What will you do about the cleft?

> Refer to a **cleft team** (e.g. CoRSU or another Smile Train partner) for free surgery: lip at about 3 months when weight and Hb allow (rule of 10s), palate at 9–12 months; arrange hearing follow-up (OME) and later speech therapy and dental care.

4. How do you assess and treat the foot?

> Rigid CAVE deformity = **structural club foot**; examine spine and hips; **Pirani score**; refer to the hospital's **Ponseti clinic** now: weekly casts (cavus first), Achilles tenotomy, then the foot-abduction brace 23 h/day for 3 months and nights to 4–5 years.

5. How do you address the relatives' beliefs and the mother's isolation?

> Explain that clefts and club foot are **common birth differences**, not curses or contagious, and both are very treatable; show before-and-after examples if available; involve the father and family; connect her to other parents and the social worker; encourage her to continue immunisations and clinic visits.

6. Give his immunisations today?

> **Yes**: give all 6-week UNEPI vaccines (OPV1, IPV1, DPT-HepB-Hib1, PCV1, rotavirus 1). Clefts and club foot are not contraindications; give the oral rotavirus vaccine slowly into the inner cheek.

Take-away: Feeding and growth come first in a baby with a cleft; club foot needs early Ponseti treatment; both families need stigma-busting counselling.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.